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Bit line pre-charge circuit and method

Abstract

A bit line is pre-charged based on a clock signal internal to a bit line pre-charge circuit when a bit line pre-charge window is within a margin of a predetermined pre-charge window. A bit line is pre-charged based on a clock signal external to the bit line pre-charge circuit when the bit line pre-charge window is outside the margin of the predetermined pre-charge window.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. Non-Provisional application Ser. No. 17/331,718, titled “BIT LINE PRE-CHARGE CIRCUIT AND METHOD” and filed on May 27, 2021, which claims priority to U.S. Provisional Application 63/163,609, titled “Apparatus for Clock-Based Bitline (BL) and Bitline Bar (BLB) Pre-Charging” and filed on Mar. 19, 2021. U.S. Non-Provisional application Ser. No. 17/331,718 and U.S. Provisional Application 63/163,609 are incorporated herein by reference.

BACKGROUND

(1) Memory cell dies are formed by patterning a wafer. A single patterned wafer may contain thousands of memory cell dies. Memory cell dies are used in a multitude of electronic devices, such as mobile phones, laptops, desktops, tablets, watches, gaming systems, and various other industrial, commercial, and consumer electronics. Memory cells of a memory cell die may be arranged in a matrix. Each memory cell is accessed for memory read, write, and/or erase operations. Accessing a memory cell may include driving word lines and bit lines coupled to the memory cell to predetermined voltage(s).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

- (2) FIG. 1 is an illustration of a bit line pre-charge circuit, according to some embodiments.
- (3) FIG. 2 is an illustration of a bit line pre-charge circuit, according to some embodiments.
- (4) FIG. 3 is a timing diagram of a bit line pre-charge circuit, according to some embodiments.
- (5) FIG. 4 is an illustration of a bit line pre-charge circuit, according to some embodiments.
- (6) FIG. 5 is a timing diagram of a bit line pre-charge circuit, according to some embodiments.
- (7) FIG. 6 is an illustration of a waveform of a tracking circuit, according to some embodiments.
- (8) FIG. 7 illustrates a memory cell, according to some embodiments.
- (9) FIG. 8 illustrates a memory cell array, according to some embodiments.
- (10) FIG. 9 is an illustration of a method to pre-charge a bit line, according to some embodiments.
- (11) FIG. 10 is an illustration of an exemplary computer-readable medium, according to some embodiments.
- (12) FIG. 11 illustrates an example computing environment wherein one or more of the provisions set forth herein may be implemented, according to some embodiments.

DETAILED DESCRIPTION

(13) The following disclosure provides several different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments or configurations discussed.

(14) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation illustrated in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. Also, relationship terms such as “connected to,” “adjacent to,” “coupled to,” and the like, may be used herein to describe both direct and indirect relationships. “Directly” connected, adjacent, or coupled may refer to a relationship in which there are no intervening components, devices, or structures. “Indirectly” connected, adjacent, or coupled may refer to a relationship in which there are intervening components, devices, or structures.

(15) A bit line pre-charge circuit and method are disclosed. A bit line is pre-charged to a predetermined voltage by the bit line pre-charge circuit. The bit line is pre-charged within a time frame defined by a bit line pre-charge window. The bit line pre-charge window includes a margin for timing variations. Timing variations may occur due to variations in the silicon of a die, as an example. A word line signal is provided by the bit line pre-charge circuit after the bit line pre-charge window closes. However, when the margin is not adequate, the word line signal could be provided before the bit line pre-charge window closes, which could cause signal and/or data errors.

(16) According to some embodiments, when the bit line pre-charge window is within an adequate margin of a predetermined bit line pre-charge window, a bit line pre-charge clock signal is based on an edge of a pulse of a clock signal internal to the bit line pre-charge circuit. When the bit line pre-charge window is outside the margin of the predetermined bit line pre-charge window, the bit line pre-charge clock signal is based on an edge of a pulse of a clock signal external to the bit line pre-charge circuit.

(17) FIG. 1 is an illustration of a bit line pre-charge circuit **100**, according to some embodiments.

The bit line pre-charge circuit **100** is coupled to an external clock CLK_EX **102** that is external of the bit line pre-charge circuit **100**, such as a system clock as an electronic device comprising the bit line pre-charge circuit **100**. The external clock CLK_EX **102** generates and outputs a first clock signal CLK at a frequency based upon a frequency signal FRQ supplied by a controller **124**. The frequency of the first clock signal CLK is adjustable by the controller **124** using the frequency signal FRQ. In some embodiments, adjusting the frequency of the first clock signal CLK comprises adjusting the frequency of the first clock signal CLK through multiple frequencies, referred to as a frequency sweep. A frequency sweep may begin at a start frequency and end at a stop frequency. The frequency sweep through the multiple frequencies may occur at a specified sweep rate and/or specified frequency step. The frequency sweep may move up or down a frequency band, with either linear or logarithmic spacing. Other techniques to adjust the frequency of the first clock signal CLK are within the scope of the present disclosure.

(18) An output terminal **104** of the external clock CLK_EX **102** is coupled to a first input terminal **106** of a first clock signal generator CLK_G1 **108**. According to some embodiments, the first clock signal generator CLK_G1 **108** receives the first clock signal CLK and generates a second clock signal CKPB_NAP. According to some embodiments, a first edge of a pulse of the first clock signal CLK triggers a first edge of a pulse of the second clock signal CKPB_NAP. A second edge of the pulse of the second clock signal CKPB_NAP may be triggered by a first edge of a pulse of a third clock signal CKPB generated by a second clock signal generator CLK_G2 **110**. An output terminal **150** of the second clock signal generator CLK_G2 **110** is coupled to a second input terminal **152** of the first clock signal generator CLK_G1 **108**. A pulse width of the pulse of the second clock signal CKPB_NAP is defined as the distance from the first edge of the pulse of the second clock signal CKPB_NAP to the second edge of the pulse of the second clock signal CKPB_NAP. Other configurations of the bit line pre-charge circuit **100** to generate the second clock signal CKPB_NAP are within the scope of the present disclosure.

(19) An output terminal **112** of the first clock signal generator CLK_G1 **108** is coupled to an input terminal **114** of a bit line pre-charge signal generator BLPREB **116**. According to some embodiments, the bit line pre-charge signal generator BLPREB **116** receives the second clock signal CKPB_NAP and generates a bit line pre-charge signal BLPREB. The bit line pre-charge signal generator BLPREB **116** comprises an output terminal **118** to output the bit line pre-charge signal BLPREB to a memory cell array and/or a memory cell (not shown). According to some embodiments, the first edge of the pulse of the second clock signal CKPB_NAP triggers a first edge of a pulse of the bit line pre-charge signal BLPREB. The second edge of the pulse of the second clock signal CKPB_NAP triggers a second edge of the pulse of the bit line pre-charge signal BLPREB. A pulse width of the pulse of the bit line pre-charge signal BLPREB is defined as the distance from the first edge of the pulse of the bit line pre-charge signal BLPREB to the second edge of the pulse of the bit line pre-charge signal BLPREB. As may be further understood with respect to FIG. 3 and the accompanying text, in some embodiments, the pulse width of the pulse of the bit line pre-charge signal BLPREB is equal to the pulse width of the pulse of the second clock signal CKPB_NAP, but the pulse of the bit line pre-charge signal BLPREB may be time-delayed relative to a pulse of the second clock signal CKPB_NAP. For example, the bit line pre-charge signal generator BLPREB **116** may introduce a time-delay to the second clock signal CKPB_NAP to generate the bit line pre-charge signal BLPREB. In some embodiments, the bit line pre-charge signal generator BLPREB **116** comprises an inverter chain configured to introduce the time-delay, although other circuit elements for introducing a time-delay are contemplated. Other configurations of the bit line pre-charge circuit **100** to generate the bit line pre-charge signal BLPREB are within the scope of the present disclosure.

(20) A tracking circuit **120** comprises an input terminal **122** coupled to the output terminal **112** of the first clock signal generator CLK_G1 **108**. According to some embodiments, the tracking circuit **120** receives the second clock signal CKPB_NAP and determines whether the pulse width of the

second clock signal CKPB_NAP is within a margin of a predetermined bit line pre-charge window and communicates the determination to the controller **124**. In some embodiments, the tracking circuit **120** further passes the second clock signal CKPB_NAP to an output terminal **138** of the tracking circuit **120** or passes a time-delayed version of the second clock signal CKPB_NAP to the output terminal **138** of the tracking circuit **120**.

(21) The controller **124** comprises a first output terminal **126** and a second output terminal **128**. The controller **124** outputs a first switch control signal TMPT to the first output terminal **126** and outputs the frequency signal FRQ to the second output terminal **128**. The first output terminal **126** is coupled to an inverter **130** and to a switch circuit **132**. The first output terminal **126** conveys the first switch control signal TMPT to the switch circuit **132** and to the inverter **130**. The inverter **130** inverts the first switch control signal TMPT to generate a second switch control signal TMPTB. The switch circuit **132** receives the first switch control signal TMPT and the second switch control signal TMPTB. The first switch control signal TMPT and the second switch control signal TMPTB control a state of the switch circuit **132**. According to some embodiments, when the first switch control signal TMPT is at a first state (e.g., logic 0) and the second switch control signal TMPTB is at a second state (e.g., logic 1), a first switch **134** of the switch circuit **132** is closed and a second switch **136** of the switch circuit **132** is open. According to some embodiments, when the first switch control signal TMPT is at the second state (e.g., logic 1) and the second switch control signal TMPTB is at the first state (e.g., logic 0), the first switch **134** of the switch circuit **132** is open and the second switch **136** of the switch circuit **132** is closed. Other configurations of the controller **124** and the switch circuit **132** are within the scope of the present disclosure.

(22) If the first switch **134** of the switch circuit **132** is closed and the second switch **136** is open, an output terminal **138** of the tracking circuit **120** is coupled through the switch circuit **132** and an output terminal **141** of the switch circuit **132** to an input terminal **140** of the second clock signal generator CLK_G2 **110**. In this state of the switch circuit **132**, the second clock signal CKPB_NAP is communicated as CLKB_NAP_IN signal to the input terminal **140** of the second clock signal generator CLK_G2 **110**.

(23) If the first switch **134** of the switch circuit **132** is open and the second switch **136** is closed, the output terminal **104** of the external clock CLK_EX **102** is coupled through the switch circuit **132** and the output terminal **141** of the switch circuit **132** to the input terminal **140** of the second clock signal generator CLK_G2 **110**. In this state of the switch circuit **132**, the first clock signal CLK is communicated as the CLKB_NAP_IN signal to the input terminal **140** of the second clock signal generator CLK_G2 **110**. Other configurations of the switch circuit **132** are within the scope of the present disclosure.

(24) Referring to the tracking circuit **120** and the controller **124**, when the pulse width of the pulse of the second clock signal CKPB_NAP is within the margin of the predetermined bit line pre-charge window, the controller **124** sets the state of the first switch control signal TMPT to the first state (e.g., logic 0). As will be further described with respect to FIGS. **2** and **3**, when the first switch control signal TMPT is in the first state, the bit line pre-charge circuit **100** is in an internal clock state. In the internal clock state, the second clock signal CKPB_NAP is communicated as the CLKB_NAP_IN signal to the input terminal **140** of the second clock signal generator CLK_G2 **110**, and the first edge of the pulse of the second clock signal CKPB_NAP (communicated as CLKB_NAP_IN) triggers the first edge of the pulse of the third clock signal CKPB.

(25) If the pulse width of the pulse of the second clock signal CKPB_NAP is outside the margin of the predetermined bit line pre-charge window, the controller **124** sets the state of the first switch control signal TMPT to the second state (e.g., logic 1). As will be described further with respect to FIGS. **4** and **5**, when the first switch control signal TMPT is in the second state (e.g., logic 1), and the bit line pre-charge circuit **100** is in an external clock state. In the external clock state, the first clock signal CLK is communicated as the CLKB_NAP_IN signal to the input terminal **140** of the second clock signal generator CLK_G2 **110**, and a second edge of the pulse of the first clock signal

CLK (communicated as CLKB_NAP_IN) triggers the first edge of the pulse of the third clock signal CKPB. Thus, the first edge of the pulse of the third clock signal CKPB is triggered by the second clock signal CKPB_NAP when the bit line pre-charge circuit **100** is in the internal clock state and is triggered by the first clock signal CLK when the bit line pre-charge circuit **100** is in an external clock state.

(26) In both the internal clock state and the external clock state, the first edge of the pulse of the third clock signal CKPB triggers the second edge of the pulse of the second clock signal CKPB_NAP, the first edge of the pulse of the second clock signal CKPB_NAP triggers the first edge of the pulse of the bit line pre-charge signal BLPREB, and the second edge of the pulse of the second clock signal CKPB_NAP triggers the second edge of the pulse of the bit line pre-charge signal BLPREB. The second edge of the pulse of the bit line pre-charge signal BLPREB closes the bit line pre-charge window.

(27) If the pulse width of the second clock signal CKPB_NAP is within the margin of the predetermined bit line pre-charge window, the bit line pre-charge circuit **100** is in the internal clock state. In the internal clock state, the first edge of the pulse of the second clock signal CKPB_NAP triggers the first edge of the pulse of the third clock signal CKPB, which in turn triggers the second edge of the pulse of the second clock signal CKPB_NAP, which in turn triggers the second edge of the pulse of the bit line pre-charge signal BLPREB and closes the bit line pre-charge window. When the pulse width of the second clock signal CKPB_NAP is outside the margin of the predetermined bit line pre-charge window, the bit line pre-charge circuit **100** is in the external clock state and the second edge of the pulse of the first clock signal CLK triggers the first edge of the pulse of the third clock signal CKPB, which in turn triggers the second edge of the pulse of the second clock signal CKPB_NAP, which in turn triggers the second edge of the pulse of the bit line pre-charge signal BLPREB and closes the bit line pre-charge window.

(28) In the internal clock state, the controller **124** outputs a first frequency signal FRQ to the external clock CLK_EX **102**, which fixes the frequency of the first clock signal CLK generated by the external clock CLK_EX **102** at a specified frequency. In the external clock state, the controller **124** outputs a second frequency signal FRQ to the external clock CLK_EX **102**. The second frequency signal may cause the external clock CLK_EX **102** to sweep the frequency of the first clock signal CLK through several frequencies. At one or more of the several frequencies, the tracking circuit **120** determines whether the pulse width of the pulse of the second clock signal CKPB_NAP is within the margin of the predetermined bit line pre-charge window and communicates the determination to the controller **124**. By way of the sweep the frequency of the first clock signal CLK through several frequencies, the controller **124** may fix the frequency of the first clock signal CLK to a frequency that provides a bit line pre-charge window that is within a margin of the predetermined bit line pre-charge window.

(29) According to some embodiments, the bit line pre-charge circuit **100** comprises a decoder DEC **142** that receives the third clock signal CKPB and outputs a word line signal WL at a second edge of the pulse of the third clock signal CKPB that follows the second edge (i.e., closure of the pre-charge window) of the pulse of the bit line pre-charge signal BLPREB. The word line signal WL is output at a decoder output terminal **144** to activate a row of memory cells in a memory cell array for a memory operation on one or more memory cells of the row.

(30) FIG. 2 is an illustration of the bit line pre-charge circuit **100**, according to some embodiments. The functionality of the bit line pre-charge circuit **100** illustrated in FIG. 2 is substantially equivalent to the functionality of the bit line pre-charge circuit **100** of FIG. 1 under the condition that the first switch control signal TMPT is in the first state (e.g., logic 0). In FIG. 2, for clarity of explanation, the switch circuit **132** is not shown but rather is represented by a conductor **148**. The output terminal **104** of the external clock CLK_EX **102** is electrically coupled to the first input terminal **106** of the first clock signal generator CLK_G1 **108**. Because the first switch control signal TMPT is in the first state, the first switch **134** (illustrated in FIG. 1) of the switch circuit **132**

is closed and the second switch **136** (illustrated in FIG. 1) of the switch circuit **132** is open. Because the first switch **134** of the switch circuit **132** is closed, the output terminal **138** of the tracking circuit **120** is electrically coupled to the input terminal **140** of the second clock signal generator CLK_G2 **110**. Because the second switch **136** of the switch circuit **132** is open, the output terminal **104** of the external clock CLK_EX **102** is not electrically coupled to the input terminal **140** of the second clock signal generator CLK_G2 **110**. FIG. 3 is a timing diagram **300** of the bit line pre-charge circuit **100** illustrated in FIG. 2, according to some embodiments.

(31) Referring to FIG. 2 and FIG. 3, when the first clock signal generator CLK_G1 **108** receives a rising edge **302** (i.e., a first edge) of the pulse of the first clock signal CLK, the first clock signal generator CLK_G1 **108** generates and outputs at the output terminal **112** a falling edge **304** (i.e., a first edge) of the pulse of the second clock signal CKPB_NAP. The falling edge **304** of the pulse of the second clock signal CKPB_NAP is received by the bit line pre-charge signal generator BLPREB **116** and the second clock signal generator CLK_G2 **110** (communicated as CLKB_NAP_IN). The bit line pre-charge signal generator BLPREB **116** generates and outputs at the output terminal **118** a falling edge **306** (i.e., a first edge) of the pulse of the bit line pre-charge signal BLPREB. The second clock signal generator CLK_G2 **110** generates and outputs at the output terminal **150** a falling edge **308** (i.e., a first edge) of the pulse of the third clock signal CKPB.

(32) The first clock signal generator CLK_G1 **108** receives the falling edge **308** of the pulse of the third clock signal CKPB and generates and outputs at the output terminal **112** a rising edge **310** (i.e., a second edge) of the pulse of the second clock signal CKPB_NAP. The bit line pre-charge signal generator BLPREB **116** receives the rising edge **310** of the pulse of the second clock signal CKPB_NAP and generates and outputs at the output terminal **118** a rising edge **312** (i.e., a second edge) of the pulse of the bit line pre-charge signal BLPREB. A distance from the falling edge **306** of the pulse of the bit line pre-charge signal BLPREB to the rising edge **312** of the pulse of the bit line pre-charge signal BLPREB defines a bit line pre-charge window **314** when the first switch control signal TMPT is in the first state.

(33) FIG. 4 is an illustration of the bit line pre-charge circuit **100**, according to some embodiments. The functionality of the bit line pre-charge circuit **100** illustrated in FIG. 4 is substantially equivalent to the functionality of the bit line pre-charge circuit **100** of FIG. 1 under the condition that the first switch control signal TMPT is in the second state (e.g., logic 1). In FIG. 4, for clarity of explanation, the switch circuit **132** is not shown but rather is represented by a conductor **154**. The output terminal **104** of the external clock CLK_EX **102** is electrically coupled to the first input terminal **106** of the first clock signal generator CLK_G1 **108**. Because the first switch control signal TMPT is in the second state, the first switch **134** (illustrated in FIG. 1) of the switch circuit **132** is open and the second switch **136** (illustrated in FIG. 1) of the switch circuit **132** is closed. Because the first switch **134** of the switch circuit **132** is open, the output terminal **138** (illustrated in FIG. 1) of the tracking circuit **120** is not electrically coupled to the input terminal **140** of the second clock signal generator CLK_G2 **110**. Because the second switch **136** of the switch circuit **132** is closed, the output terminal **104** of the external clock CLK_EX **102** is electrically coupled to the input terminal **140** of the second clock signal generator CLK_G2 **110**. FIG. 5 is a timing diagram **500** of the bit line pre-charge circuit **100** illustrated in FIG. 4, according to some embodiments.

(34) Referring to FIG. 4 and FIG. 5, when the first clock signal generator CLK_G1 **108** receives a rising edge **502** (i.e., a first edge) of the pulse of the first clock signal CLK, the first clock signal generator CLK_G1 **108** generates and outputs at the output terminal **112** a falling edge **504** (i.e., a first edge) of the pulse of the second clock signal CKPB_NAP. The falling edge **504** of the pulse of the second clock signal CKPB_NAP is received by the bit line pre-charge signal generator BLPREB **116**. The bit line pre-charge signal generator BLPREB **116** generates and outputs at the output terminal **118** a falling edge **506** (i.e., a first edge) of the pulse of the bit line pre-charge

signal BLPREB. When the second clock signal generator CLK_G2 **110** receives a falling edge **508** (i.e., a second edge) of the pulse of the first clock signal CLK, the second clock signal generator CLK_G2 **110** generates and outputs at the output terminal **150** a falling edge **510** (i.e., a first edge) of the pulse of the third clock signal CKPB.

(35) The first clock signal generator CLK_G1 **108** receives the falling edge **510** of the pulse of the third clock signal CKPB and generates and outputs at the output terminal **112** a rising edge **512** (i.e., a second edge) of the pulse of the second clock signal CKPB_NAP. The bit line pre-charge signal generator BLPREB **116** receives the rising edge **512** of the pulse of the second clock signal CKPB_NAP and generates and outputs at the output terminal **118** a rising edge **514** (i.e., a second edge) of the pulse of the bit line pre-charge signal BLPREB. A distance from the falling edge **506** of the pulse of the bit line pre-charge signal BLPREB to the rising edge **514** of the pulse of the bit line pre-charge signal BLPREB defines a bit line pre-charge window **516** when the first switch control signal TMPT is in the second state.

(36) FIG. **6** is an illustration of the waveform **600** of the second clock signal CKPB_NAP, according to some embodiments. The second clock signal CKPB_NAP includes a pre-charge window T_PRE **602** from t0 to t1. Because the falling **604** and rising **606** edges of the pulse of the second clock signal CKPB_NAP directly trigger the falling and rising edges of the pulse of the bit line pre-charge signal BLPREB, the duration of the pre-charge window T_PRE **602** from t0 to t1 is substantially equal to the bit line pre-charge window **314** when the first switch control signal TMPT is in the first state, and to the bit line pre-charge window **516** when the first switch control signal TMPT is in the second state. Because the tracking circuit **120** receives the second clock signal CKPB_NAP, the tracking circuit **120** receives the pre-charge window T_PRE **602**. Because the tracking circuit **120** receives the pre-charge window T_PRE **602**, the tracking circuit **120** readily determines the bit line pre-charge window **314** when the first switch control signal TMPT is in the first state or the bit line pre-charge window **516** when the first switch control signal TMPT is in the second state. According to some embodiments, the duration of the pre-charge window T_PRE **602** is communicated to the controller **124**. Responsive to the duration of the pre-charge window T_PRE **602**, the controller controls the first switch control signal TMPT to be in the first state when the duration of the pre-charge window T_PRE **602** is within the margin of a predetermined bit line pre-charge window, or controls the first switch control signal TMPT to be in the second state when the duration of the pre-charge window T_PRE **602** is outside the margin of the predetermined bit line pre-charge window.

(37) FIG. **7** illustrates a memory cell **700**, according to some embodiments. The memory cell **700** is coupled to the bit line pre-charge circuit **100**. The memory cell **700** is coupled to a bit line BL **702** and an inverted bit line BLB **704**, which is set to a level opposite to that of the bit line BL **702**. The memory cell **700** is also coupled to a word line WL **706** that is set to a first state (e.g., a high logic level) or a second state (e.g., a low logic level) according to an address signal decoded by the decoder DEC **142**. The memory cell **700** comprises a first inverter INV1 **708** and a second inverter INV2 **710**, which are cross-coupled. The memory cell **700** also comprises a first data transfer transistor T1 **712** and a second data transfer transistor T2 **714** respectively coupled to the bit line BL **702** and to the inverted bit line BLB **704**. An output of the first inverter INV1 **708** is coupled to second data transfer transistor T2 **714**, and an output of the second inverter INV2 **710** is coupled to the second data transfer transistor T1 **712**. The word line WL **706** is coupled to a gate **716** of first data transfer transistor T1 **712** and to a gate **718** of second data transfer transistor T2 **714**. When the word line WL **706** is set to the first state, the first data transfer transistor T1 **712** and second data transfer transistor T2 **714** are closed. When the word line WL **706** is set to the second state, the first data transfer transistor T1 **712** and second data transfer transistor T2 **714** are open.

(38) According to some embodiments, the decoder output terminal **144** of the decoder DEC **142** is coupled to the word line WL **706**, and the output terminal **118** of the bit line pre-charge signal generator BLPREB **116** is coupled to the inverted bit line BLB **704**. Other configurations of the

memory cell **700** and connections to the bit line pre-charge circuit **100** are within the scope of the present disclosure.

(39) FIG. **8** illustrates a memory cell array **800** comprising rows and columns of the memory cells **700**, in which each row is coupled to a different word line WL **706** and each column is coupled to a different bit line BL **702** and inverted bit line BLB **704** according to some embodiments of the present disclosure. The word lines WL **706** of the memory cell array **800** are coupled to a word line address decoder **802** that is external to the bit line pre-charge circuit **100**. The inverted bit lines BLB **704** of the memory cell array **800** are coupled to a data line decoder **804**. The data line decoder **804** decodes a data address and couples the output terminal **118** of the bit line pre-charge signal generator BLPREB **116** to a corresponding inverted bit line BLB **704**. Other configurations of the memory cell array **800** and connections to the bit line pre-charge circuit **100** are within the scope of the present disclosure.

(40) FIG. **9** is an illustration of a method **900** to pre-charge a bit line, according to some embodiments. The method **900** comprises determining whether a bit line pre-charge window **314/516** of a bit line pre-charge circuit **100** is within a margin of a predetermined bit line pre-charge window at **902**. When the bit line pre-charge window **314/516** of the bit line pre-charge circuit **100** is within the margin of the predetermined bit line pre-charge window, an edge of a pulse of a bit line pre-charge clock signal (e.g., the third clock signal CKPB) is triggered based on an edge of a pulse of a clock signal (e.g., the second clock signal CKPB_NAP) internal to the bit line pre-charge circuit **100** at **904**. When the bit line pre-charge window **314/516** of the bit line pre-charge circuit **100** is outside the margin of the predetermined bit line pre-charge window, the edge of the pulse of the bit line pre-charge clock signal (e.g., the third clock signal CKPB) is triggered based on an edge of a pulse of a clock signal (e.g., the first clock signal CLK) external to the bit line pre-charge circuit **100** at **906**. When the bit line pre-charge window **314/516** of the bit line pre-charge circuit **100** is outside the margin of the predetermined bit line pre-charge window, sweeping the frequency of the external clock signal (e.g., the first clock signal CLK) determines an external clock frequency that provides a bit line pre-charge window **314/516** that is within a margin of a predetermined bit line pre-charge window at **908**. Other and/or additional acts of the method **900** are within the scope of the present disclosure.

(41) FIG. **10** illustrates an exemplary computer-readable medium, according to some embodiments. One or more embodiments involve a computer-readable medium comprising processor-executable instructions configured to implement one or more of the techniques presented herein. An exemplary computer-readable medium is illustrated in FIG. **10**, wherein the embodiment **1000** comprises a computer-readable medium **1002** (e.g., a CD-R, DVD-R, flash drive, a platter of a hard disk drive, etc.), on which is encoded computer-readable data **1004**. This computer-readable data **1004** in turn comprises a set of processor-executable computer instructions **1006** that when executed are configured to facilitate operations according to one or more of the principles set forth herein. In some embodiments **1000**, the processor-executable computer instructions **1006** are configured to facilitate performance of a method **1008**, such as at least some of the aforementioned method **900**. In some embodiments, the processor-executable computer instructions **1006** are configured to facilitate implementation of a system, such as at least some of the one or more aforementioned system(s). Many such computer-readable media may be devised by those of ordinary skill in the art that are configured to operate in accordance with the techniques presented herein.

(42) FIG. **11** illustrates an example computing environment wherein one or more of the provisions set forth herein may be implemented, according to some embodiments. FIG. **11** and the following discussion provide a brief, general description of a suitable computing environment to implement embodiments of one or more of the provisions set forth herein. The computing environment of FIG. **11** is only one example of a suitable computing environment and is not intended to suggest any limitation as to the scope of use or functionality of the computing environment. Example computing devices include, but are not limited to, personal computers, server computers, hand-held

or laptop devices, mobile devices (such as mobile phones, Personal Digital Assistants (PDAs), media players, and the like), multiprocessor systems, consumer electronics, mini computers, mainframe computers, distributed computing environments that include any of the above systems or devices, and the like.

(43) Although not required, embodiments are described in the general context of “computer readable instructions” being executed by one or more computing devices. Computer readable instructions may be distributed via computer readable media (discussed below). Computer readable instructions may be implemented as program modules, such as functions, objects, Application Programming Interfaces (APIs), data structures, and the like, that perform particular tasks or implement particular abstract data types. Typically, the functionality of the computer readable instructions may be combined or distributed as desired in various environments.

(44) FIG. 11 depicts an example of a system 1100 comprising a computing device 1102 configured as a controller to implement embodiments provided herein. In some configurations, the computing device 1102 includes at least one processing unit 1106 and memory 1108. Depending on the exact configuration and type of computing device, memory 1108 may be volatile (such as random-access memory (RAM), for example), non-volatile (such as read-only memory (ROM), flash memory, etc., for example), or some combination of the two. This configuration is illustrated in FIG. 11 by dashed line 1104.

(45) In some embodiments, the computing device 1102 may include additional features and/or functionality. For example, computing device 1102 may also include additional storage (e.g., removable and/or non-removable) including, but not limited to, magnetic storage, optical storage, and the like. Such additional storage is illustrated in FIG. 11 by storage 1110. In some embodiments, computer readable instructions to implement one or more embodiments provided herein may be in the storage 1110. The storage 1110 may also store other computer readable instructions to implement an operating system, an application program, and the like. Computer readable instructions may be loaded in the memory 1108 for execution by the processing unit 1106, for example.

(46) The term “computer readable media” as used herein includes computer storage media. Computer storage media includes volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information such as computer readable instructions or other data. The memory 1108 and the storage 1110 are examples of computer storage media. Computer storage media includes, but is not limited to, RAM, ROM, electrically erasable programmable read-only memory (EEPROM), flash memory, or other memory technology, CD-ROM, Digital Versatile Disks (DVDs), or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage, or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by the computing device 1102. Any such computer storage media may be part of the computing device 1102.

(47) The computing device 1102 may also include communication connection(s) 1116 that allows the computing device 1102 to communicate with other devices. The communication connection(s) 1116 may include, but is not limited to, a modem, a Network Interface Card (NIC), an integrated network interface, a radio frequency transmitter/receiver, an infrared port, a universal serial bus (USB) connection, or other interfaces for connecting the computing device 1102 to other computing devices. The communication connection(s) 1116 may include a wired connection or a wireless connection. The communication connection(s) 1116 may transmit and/or receive communication media.

(48) The term “computer readable media” may include communication media. Communication media typically embodies computer readable instructions or other data in a “modulated data signal” such as a carrier wave or other transport mechanism and includes any information delivery media. The term “modulated data signal” may include a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal.

(49) The computing device **1102** may include input device(s) **1114** such as keyboard, mouse, pen, voice input device, touch input device, infrared cameras, video input devices, and/or any other input device. Output device(s) **1112** such as one or more displays, speakers, printers, and/or any other output device may also be included in the computing device **1102**. The input device(s) **1114** and the output device(s) **1112** may be connected to the computing device **1102** via a wired connection, wireless connection, or any combination thereof. In some embodiments, an input device or an output device from another computing device may be used as the input device(s) **1114** or the output device(s) **1112** for the computing device **1102**.

(50) Components of the computing device **1102** may be connected by various interconnects, such as a bus. Such interconnects may include a Peripheral Component Interconnect (PCI), such as PCI Express, USB, firewire (IEEE 1394), an optical bus structure, and the like. In some embodiments, components of the computing device **1102** may be interconnected by a network. For example, the memory **1108** may be comprised of multiple physical memory units located in different physical locations interconnected by a network.

(51) Those skilled in the art will realize that storage devices utilized to store computer readable instructions may be distributed across a network. For example, a second computing device **1120** accessible via a network **1118** may store computer readable instructions to implement one or more embodiments provided herein. The computing device **1102** may access the second computing device **1120** and download a part or all of the computer readable instructions for execution. Alternatively, the computing device **1102** may download pieces of the computer readable instructions, as needed, or some instructions may be executed at the computing device **1102** and some at the second computing device **1120**.

(52) As disclosed, the bit line pre-charge signal generator BLPREB **116** comprises an output terminal **118**. The output terminal **118** is coupled to a bit line (e.g., the bit line BL **702** or inverted bit line BLB **704**) of the memory cell array **800**. The bit line is pre-charged to the predetermined voltage by the bit line pre-charge circuit **100**. The bit line is pre-charged within the time frame defined by the bit line pre-charge window **314/516**. The bit line pre-charge window **314/516** includes a margin for timing variations. The word line signal WL is provided by the bit line pre-charge circuit **100** after the bit line pre-charge window **314/516** closes. When the bit line pre-charge window **314/516** is within an adequate margin of a predetermined bit line pre-charge window, the bit line pre-charge clock signal is based on an edge of a pulse of the clock signal CKPB_NAP internal to the bit line pre-charge circuit. When the bit line pre-charge window **314/516** is outside the margin of the predetermined bit line pre-charge window, the bit line pre-charge clock signal is based on an edge of a pulse of a clock signal CLK external to the bit line pre-charge circuit **100**.

(53) According to some embodiments, a bit line pre-charge circuit includes a bit line pre-charge signal generator, a first clock signal generator coupled to the bit line pre-charge signal generator and including an output terminal, a second clock signal generator coupled to the first clock signal generator and including an input terminal, and a switch circuit including an output terminal coupled to the input terminal of the second clock signal generator. The first clock signal generator is coupled to the switch circuit. When the switch circuit is in a first state, the input terminal of the second clock signal generator is electrically coupled to an external clock, external to the bit line pre-charge circuit. When the switch circuit is in a second state, the output terminal of the first clock signal generator is coupled to the input terminal of the second clock signal generator.

(54) According to some embodiments, a bit line pre-charge circuit includes a bit line pre-charge signal generator including an input terminal, a first clock signal generator including an input terminal and an output terminal, a second clock signal generator including an input terminal, and a switch circuit including an output terminal. The input terminal of the bit line pre-charge signal generator is electrically coupled to the output terminal of the first clock signal generator and to the input terminal of the second clock signal generator, the output terminal of the switch circuit is electrically coupled to the input terminal of the second clock signal generator, and the input

terminal of the first clock signal generator is electrically coupled to an output terminal of an external clock external to the bit line pre-charge circuit.

(55) According to some embodiments, a method to pre-charge a bit line includes determining whether a bit line pre-charge window of a bit line pre-charge circuit is within a margin of a predetermined bit line pre-charge window, when the bit line pre-charge window of the bit line pre-charge circuit is within the margin of the predetermined bit line pre-charge window, triggering an edge of a pulse of a bit line pre-charge clock signal based on an edge of a pulse of a clock signal internal to the bit line pre-charge circuit, and when the bit line pre-charge window of the bit line pre-charge circuit is outside the margin of the predetermined bit line pre-charge window, triggering the edge of the pulse of the bit line pre-charge clock signal based on an edge of a pulse of a clock signal external to the bit line pre-charge circuit.

(56) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

(57) Although the subject matter has been described in language specific to structural features or methodological acts, it is to be understood that the subject matter of the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing at least some of the claims.

(58) Various operations of embodiments are provided herein. The order in which some or all of the operations are described should not be construed to imply that these operations are necessarily order dependent. Alternative ordering will be appreciated having the benefit of this description. Further, it will be understood that not all operations are necessarily present in each embodiment provided herein. Also, it will be understood that not all operations are necessary in some embodiments.

(59) It will be appreciated that layers, features, elements, etc. depicted herein are illustrated with particular dimensions relative to one another, such as structural dimensions or orientations, for example, for purposes of simplicity and ease of understanding and that actual dimensions of the same differ substantially from that illustrated herein, in some embodiments. Additionally, a variety of techniques exist for forming the layers, regions, features, elements, etc. mentioned herein, such as at least one of etching techniques, planarization techniques, implanting techniques, doping techniques, spin-on techniques, sputtering techniques, growth techniques, or deposition techniques such as CVD, for example.

(60) Moreover, “exemplary” is used herein to mean serving as an example, instance, illustration, etc., and not necessarily as advantageous. As used in this application, “or” is intended to mean an inclusive “or” rather than an exclusive “or”. In addition, “a” and “an” as used in this application and the appended claims are generally to be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form. Also, at least one of A and B and/or the like generally means A or B or both A and B. Furthermore, to the extent that “includes”, “having”, “has”, “with”, or variants thereof are used, such terms are intended to be inclusive in a manner similar to the term “comprising”. Also, unless specified otherwise, “first,” “second,” or the like are not intended to imply a temporal aspect, a spatial aspect, an ordering, etc. Rather, such terms are merely used as identifiers, names, etc. for features, elements, items, etc. For example, a first element and a second element generally correspond to element A and element B or two different or two identical elements or the same element.

(61) Also, although the disclosure has been shown and described with respect to one or more

implementations, equivalent alterations and modifications will occur to others of ordinary skill in the art based upon a reading and understanding of this specification and the annexed drawings. The disclosure comprises all such modifications and alterations and is limited only by the scope of the following claims. In particular regard to the various functions performed by the above described components (e.g., elements, resources, etc.), the terms used to describe such components are intended to correspond, unless otherwise indicated, to any component which performs the specified function of the described component (e.g., that is functionally equivalent), even though not structurally equivalent to the disclosed structure. In addition, while a particular feature of the disclosure may have been disclosed with respect to only one of several implementations, such feature may be combined with one or more other features of the other implementations as may be desired and advantageous for any given or particular application.

Claims

1. A method, comprising: when a switch circuit is in a first state, coupling an input terminal of a first clock signal generator to an external clock through the switch circuit and supplying an external clock signal generated by the external clock to the first clock signal generator and a second clock signal generator; and when the switch circuit is in a second state, coupling an output terminal of the second clock signal generator to the input terminal of first clock signal generator while continuing to supply the external clock signal to the second clock signal generator.
2. The method of claim 1, wherein coupling the output terminal of the second clock signal generator to the input terminal of first clock signal generator comprises: coupling the output terminal of the second clock signal generator to the input terminal of first clock signal generator through a tracking circuit.
3. The method of claim 2, wherein: a first clock signal is output at the output terminal of the second clock signal generator, and the tracking circuit delays the first clock signal such that a time-delayed version of the first clock signal is supplied to the input terminal of first clock signal generator when the switch circuit is in the second state.
4. The method of claim 1, wherein: a first clock signal is output at the output terminal of the second clock signal generator, and the first clock signal is output to a bit line pre-charge signal generator configured to generate a bit line pre-charge signal based on the first clock signal.
5. The method of claim 4, wherein a pulse width of the bit line pre-charge signal is equal to a pulse width of the first clock signal.
6. The method of claim 1, wherein: a first clock signal is output at the output terminal of the second clock signal generator, and the method comprises: controlling the switch circuit to be in the first state when a pulse width of the first clock signal is outside a margin of a predetermined bit line pre-charge window.
7. The method of claim 6, comprising: controlling the external clock to perform a frequency sweep to vary a frequency of the external clock signal when the first clock signal is outside the margin of the predetermined bit line pre-charge window.
8. The method of claim 6, comprising: controlling the switch circuit to be in the second state when the pulse width of the first clock signal is within the margin of the predetermined bit line pre-charge window.
9. The method of claim 8, comprising: controlling the external clock to maintain a frequency of the external clock signal at a fixed frequency when the first clock signal is within the margin of the predetermined bit line pre-charge window.
10. The method of claim 9, comprising: controlling the external clock to perform a frequency sweep to vary the frequency of the external clock signal when the first clock signal is outside the margin of the predetermined bit line pre-charge window.
11. The method of claim 1, wherein: a first clock signal is output at the output terminal of the

- second clock signal generator, a timing of a first edge of a pulse of the first clock signal is a function of a second edge of the pulse of the first clock signal when the switch circuit is in the second state, and the timing of the first edge of the pulse of the first clock signal is a function of an edge of a pulse of the external clock signal when the switch circuit is in the first state.
12. A bit line pre-charge circuit, comprising: a first clock signal generator comprising an output terminal; a second clock signal generator coupled to the first clock signal generator and comprising an input terminal; and a switch circuit comprising an output terminal coupled to the input terminal of the second clock signal generator, wherein: when the switch circuit is in a first state, the input terminal of the second clock signal generator is coupled to an external clock through the switch circuit, and when the switch circuit is in a second state, the output terminal of the first clock signal generator is coupled to the input terminal of the second clock signal generator through the switch circuit.
13. The bit line pre-charge circuit of claim 12, comprising: a tracking circuit disposed between the output terminal of the first clock signal generator and the input terminal of the second clock signal generator.
14. The bit line pre-charge circuit of claim 13, wherein the tracking circuit determines whether a pulse width of a first clock signal generated by the first clock signal generator is within a margin of a predetermined bit line pre-charge window to yield a determination.
15. The bit line pre-charge circuit of claim 14, comprising: a controller configured to receive the determination from the tracking circuit and to place the switch circuit in the first state or the second state based upon the determination.
16. The bit line pre-charge circuit of claim 15, wherein: the controller places the switch circuit in the first state when the pulse width of the first clock signal is outside the margin of the predetermined bit line pre-charge window, and the controller places the switch circuit in the second state when the pulse width of the first clock signal is within the margin of the predetermined bit line pre-charge window.
17. The bit line pre-charge circuit of claim 12, wherein an output terminal of the second clock signal generator is coupled to an input terminal of the first clock signal generator.
18. A bit line pre-charge circuit, comprising: a first clock signal generator configured to output a first clock signal, and a switch circuit comprising an output terminal configured to output the first clock signal when the switch circuit is in a first state or an external clock signal generated by an external clock when the switch circuit is in a second state, wherein: a timing of a first edge of a pulse of the first clock signal is a function of a second edge of the pulse of the first clock signal when the switch circuit is in the first state, and the timing of the first edge of the pulse of the first clock signal is a function of an edge of a pulse of the external clock signal when the switch circuit is in the second state.
19. The bit line pre-charge circuit of claim 18, comprising: a controller configured to place the switch circuit in the first state or in the second state as a function of a pulse width of the first clock signal.
20. The bit line pre-charge circuit of claim 18, comprising: a second clock signal generator comprising an output terminal coupled to an input terminal of the first clock signal.
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